

When More Functions Cannot Help: Moment Obstructions and Exact Degree Gaps for Discrete-Time Safety Certificates

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Research question. At a prescribed polynomial degree, is failure of safety-certificate synthesis caused by insufficient functions, or by the global constant-comparison condition itself? Can the latter be diagnosed without choosing the number of functions or the comparison matrix?

Answer developed here. Constant comparisons are equivalent, under explicit regularity assumptions, to preservation of the nonpositive orthant by a *convexified certificate-value transition relation*. A truncated-moment witness then rules out every fixed-anchor VBC at a specified degree. An exact polynomial example has minimum implication-IBC degree two and minimum constant-comparison fixed-anchor VBC degree four, allowing arbitrary finite component counts and either propagation direction.

1 Definitions and a characterization for a fixed vector

Retain the submitted paper’s autonomous system $x^+ = f(x)$, nonempty disjoint $X_0, X_u \subseteq X \subseteq \mathbb{R}^n$, and $f(X) \subseteq X$. A forward strict-margin fixed-anchor VBC is $B = (B_1, \dots, B_m)^\top$ and a constant entrywise nonnegative A satisfying

$$B \leq -\delta_0 \mathbf{1} \text{ on } X_0, \quad B_1 \geq \delta_u \text{ on } X_u, \quad B(f(x)) \leq AB(x) \text{ on } X, \quad (1)$$

where $\delta_0, \delta_u > 0$. The backward definition reverses the separation sets and uses $B(x) \leq AB(f(x))$. A forward implication IBC has frames b_i , $b_1 \leq 0$ on X_0 , every $b_i > 0$ on X_u , and $b_i(x) \leq 0 \Rightarrow b_{\min\{i+1, m\}}(f(x)) \leq 0$ on X . A single frame is permitted. No bound on $\rho(A)$ is imposed. All degree bounds are total polynomial degree, including constants.

Proposed theorem 1 (Global comparison versus convexified invariance). *Assume additionally that X is compact, f and B are continuous, and some $x_* \in X$ satisfies $B(x_*) < 0$ componentwise. Define*

$$Q_B = \text{conv}\{(B(x), B(f(x))) : x \in X\} \subseteq \mathbb{R}^{2m}.$$

The following are equivalent:

1. *There exists a constant $A \geq 0$ with $B(f(x)) \leq AB(x)$ throughout X .*
2. *Every $(s, t) \in Q_B$ with $s \leq 0$ satisfies $t \leq 0$.*
3. *For every probability measure μ on X , $\int B \, d\mu \leq 0$ implies $\int B \circ f \, d\mu \leq 0$.*

If no such A exists, a violating measure can be chosen with at most $m + 2$ atoms.

Proof. The inequality $t \leq As$ is affine and therefore persists on Q_B ; $A \geq 0$ gives (1) $\Rightarrow$ (2). Compactness makes Q_B compact, and its elements are exactly the barycenters of the transition-value graph, giving (2) $\Leftrightarrow$ (3). For each row i , maximize t_i over Q_B subject to $s \leq 0$. Under (2) its value $p_i \leq 0$. The point induced by x_* is strictly feasible; mixing it slightly with a relative-interior point gives strict feasibility in the affine hull. Finite-dimensional convex duality, applied on the affine hull of Q_B , gives an attained multiplier $a_i \geq 0$ with

$$\max_{(s, t) \in Q_B} (t_i - a_i^\top s) = p_i \leq 0.$$

These row multipliers form A . If (2) fails in row i , apply Caratheodory’s theorem to the projected graph $(B(x), B_i(f(x))) \in \mathbb{R}^{m+1}$ to obtain at most $m + 2$ atoms. $\square$

2 A degree obstruction independent of component count

Theorem 1 distinguishes ordinary invariance of $S = \{B \leq 0\}$ from the extra requirement: preservation of the orthant after mixing *transition values*, even if the mixture is not induced by one state. It supplies the following useful sufficient regime. If X is convex, every B_j is convex and every $B_i \circ f$ is concave, ordinary invariance implies condition (3): Jensen’s inequalities give $B(\bar{x}) \leq \int B \, d\mu \leq 0$ and $\int B_i \circ f \, d\mu \leq B_i(f(\bar{x})) \leq 0$. With the theorem’s other assumptions, a constant comparison matrix therefore exists. These are sufficient conditions, not necessary ones.

Let $v_d(x)$ contain all monomials of total degree at most d , including 1; its dimension is $N = \binom{n+d}{d}$. Write $M_d(\mu) = \int v_d \, d\mu$.

Proposed theorem 2 (Truncated-moment obstruction). *Suppose there exist finitely supported probability measures μ on X , μ_0 on X_0 , and ν on X_u such that*

$$M_d(\mu) = M_d(\mu_0), \quad M_d(f_{\#}\mu) = M_d(\nu). \quad (2)$$

Then no fixed-anchor polynomial VBC of degree at most d exists, in either propagation direction, for any finite $m \geq 1$ or any constant $A \geq 0$.

Proof. For every degree- d vector B , the moment identities imply $\int B \, d\mu = \int B \, d\mu_0$ and $\int B \circ f \, d\mu = \int B \, d\nu$. Integrating the forward comparison gives

$$\int B \, d\nu \leq A \int B \, d\mu_0 \leq 0,$$

contradicting $\int B_1 \, d\nu \geq \delta_u > 0$. For the backward definition, integration yields $\int B \, d\mu_0 \leq A \int B \, d\nu \leq 0$, contradicting its positive initial anchor. Neither argument restricts m or the entries of A beyond nonnegativity. $\square$

Geometric interpretation. Define $K_{0,d} = \text{conv } v_d(X_0)$, $K_{u,d} = \text{conv } v_d(X_u)$ and

$$F_d = \text{conv}\{(v_d(x), v_d(f(x))) : x \in X\}.$$

The witness says $F_d \cap (K_{0,d} \times K_{u,d}) \neq \emptyset$. For $d = 1$, point-mass endpoint measures recover the affine convex-combination obstruction. In the supplied submission it is *Theorem 9*; item 10 is a scope remark. For compact continuous data, Caratheodory bounds the required atom counts by $2N - 1$ for μ and N each for μ_0, ν .

Existence interpretation. Under Theorem 1’s compactness assumptions, degree- d forward VBC existence is equivalent to existence of a finite matrix C whose feature polyhedron $\{z : Cz \leq 0\}$ has strict initial inclusion, a fixed unsafe separating row, and is invariant under the relation F_d . Indeed, $B = Cv_d$ and linear projection commute with convexification, so Theorem 1 supplies A . More rows refine the feature polyhedron; they cannot remove transitions already in F_d .

Limits. This is a sufficient nonexistence test, not an iff criterion for all unknown vectors. It uses the fixed-anchor condition. A mixture of unsafe states need not violate the general pointwise-separation VBC definition. Also, for $d \geq 2$, matching the moments of a *single* initial point forces zero variance and μ to be that point mass. Distributed initial endpoints are essential in the example below.

3 Exact example: quadratic implications, quartic comparisons

Set

$$\begin{aligned} f(x) &= x \left[1 + \frac{1}{2} \left(x^2 - \frac{1}{4} \right) (1 - x^2) \right] = \frac{7}{8}x + \frac{5}{8}x^3 - \frac{1}{2}x^5, & X &= [-1, 1], \\ a &= \frac{12}{25}, \quad u = \frac{1641}{3200}, \quad v = \frac{103}{200}, & X_0 &= [-a, a], \quad X_u = [-v, -u] \cup [u, v]. \end{aligned}$$

These are compact semialgebraic sets and $a < 1/2 < u < v < 1$.

Invariant verification domain. With $z = x^2 \in [0, 1]$,

$$f'(x) = \frac{1}{4} + (1 - z) \left(\frac{5}{8} + \frac{5}{2}z \right) \geq \frac{1}{4}.$$

Since $f(\pm 1) = \pm 1$, monotonicity proves $f(X) \subseteq X$. Write $f(x) = xh(z)$, where $h(z) = 1 + \frac{1}{2}(z - \frac{1}{4})(1 - z) \geq 7/8 > 0$.

Explicit implication IBC. Take the single quadratic frame

$$b(x) = x^2 - \frac{1}{4}.$$

If $b(x) \leq 0$, then $0 < h(x^2) \leq 1$, hence $|f(x)| \leq |x| \leq 1/2$. This is the required implication on all X . The separation margins are

$$b \leq -\frac{49}{2500} \text{ on } X_0, \quad b \geq \frac{132881}{10240000} \text{ on } X_u.$$

This is a single-frame IBC, also a scalar sign-inductive barrier; it isolates the propagation-rule gap, not a benefit from adding frames. An affine forward IBC is impossible: its first frame would be positive at both u and $-u$ and thus positive at $0 \in X_0$. In the backward case, its first frame is nonpositive at $\pm u$, hence nonpositive at 0 , contradicting required positivity on X_0 . Therefore the minimum implication-IBC degree is exactly two, even allowing arbitrarily many frames and either direction.

All quadratic and cubic fixed-anchor VBCs are impossible. Let $t = 3/4$ and $\theta = (a/t)^2 = 256/625$. Then $f(t) = 1641/2048$ and $u = af(t)/t$. Define

$$\begin{aligned} \mu &= \frac{369}{625}\delta_0 + \frac{128}{625}(\delta_{-t} + \delta_t), \\ \mu_0 &= \frac{1}{2}(\delta_{-a} + \delta_a), \quad \nu = \frac{1}{2}(\delta_{-u} + \delta_u). \end{aligned}$$

All odd moments through degree three vanish, and

$$M_3(\mu) = M_3(\mu_0) = (1, 0, a^2, 0), \quad M_3(f_{\#}\mu) = M_3(\nu) = (1, 0, u^2, 0).$$

Theorem 2 excludes *every* degree-at-most-three fixed-anchor VBC in either direction, with no bound on m and no sparsity restriction on A . In particular, adding quadratic components or allowing cyclic coupling cannot resolve this obstruction. The same measure also disproves constant comparison for the specified $B = b$: $\int b \, d\mu = a^2 - 1/4 < 0$ but $\int b \circ f \, d\mu = u^2 - 1/4 > 0$, despite invariance of $\{b \leq 0\}$.

This is not a true unsafe trajectory. The transition-source measure places mass at $\pm 3/4$, outside X_0 . Its low-order moments impersonate an initial distribution; its successor moments impersonate an unsafe distribution. Global degree-limited comparisons cannot distinguish these distributions. The implication certificate only constrains the actual antecedent set.

4 Exact recovery and a parameterized separation family

Quartic recovery. Set

$$p(x) = \frac{3}{10000} - (x^2 - \frac{1}{4})^2, \quad A = [1].$$

Writing $z = x^2$, $h = 7/8 + 5z/8 - z^2/2$, define

$$Q(z) = 2 + \frac{15}{16}z - \frac{5}{8}z^2 - \frac{9}{16}z^3 + \frac{1}{4}z^4.$$

Exact polynomial division gives $f(x)^2 + x^2 - 1/2 = (z - 1/4)Q(z)$, hence

$$p(x) - p(f(x)) = \frac{1}{2}z(z - \frac{1}{4})^2(1 - z)(h(z) + 1)Q(z) \geq 0 \quad (x \in X). \quad (3)$$

The degree-four Bernstein coefficients of Q on $[0, 1]$ are $(2, 143/64, 227/96, 9/4, 2)$; all factors in (3) are nonnegative. The uniform margins are

$$p \leq -\frac{263}{3125000} \text{ on } X_0, \quad p \geq \frac{109119}{1600000000} \text{ on } X_u.$$

Thus this quartic scalar VBC is valid and the minimum VBC degree is exactly four.

Proposed theorem 3 (An open parameter region with the same degree gap). *Choose $0 < r < 1$ and $0 < c < \min\{1/r^2, 1/[2(1 - r^2)]\}$. Define*

$$f_{r,c}(x) = x[1 + c(x^2 - r^2)(1 - x^2)], \quad X = [-1, 1].$$

Choose $t \in (r, 1)$, put $k = f_{r,c}(t)/t > 1$, and select

$$\frac{r}{k} < a < r\sqrt{\frac{2}{k^2 + 1}}, \quad u = ak, \quad u < v < \min\{1, \sqrt{2r^2 - a^2}\}.$$

Let $X_0 = [-a, a]$, $X_u = [-v, -u] \cup [u, v]$ and choose

$$(v^2 - r^2)^2 < \kappa < (r^2 - a^2)^2.$$

The minimum implication-IBC degree is two and the minimum fixed-anchor constant-comparison VBC degree is four, allowing arbitrary finite component counts and either propagation direction. Explicit forward witnesses are $b = x^2 - r^2$ and $p = \kappa - (x^2 - r^2)^2$.

Proof. For $z \in [0, 1]$, $f'_{r,c} = 1 - cr^2 + 3c(1 + r^2)z - 5cz^2$ is concave in z with positive endpoint values $1 - cr^2$ and $1 - 2c(1 - r^2)$. Hence f is strictly increasing, $f(X) \subseteq X$, and its positive multiplier h satisfies $h \leq 1$ for $|x| \leq r$ and $h \geq 1$ for $|x| \geq r$. This proves the quadratic implication and shows that $|f(x)^2 - r^2| \geq |x^2 - r^2|$, proving the quartic global inequality. The parameter bounds give strict margins and are nonempty because $r/k < r\sqrt{2/(k^2 + 1)} < r$ for $k > 1$. The three-atom source measure with $\theta = a^2/t^2$ and symmetric endpoint measures reproduces moments through degree three. Apply Theorem 2 and the affine IBC argument from Section 3. $\square$

5 Geometry, verified computation, and related work

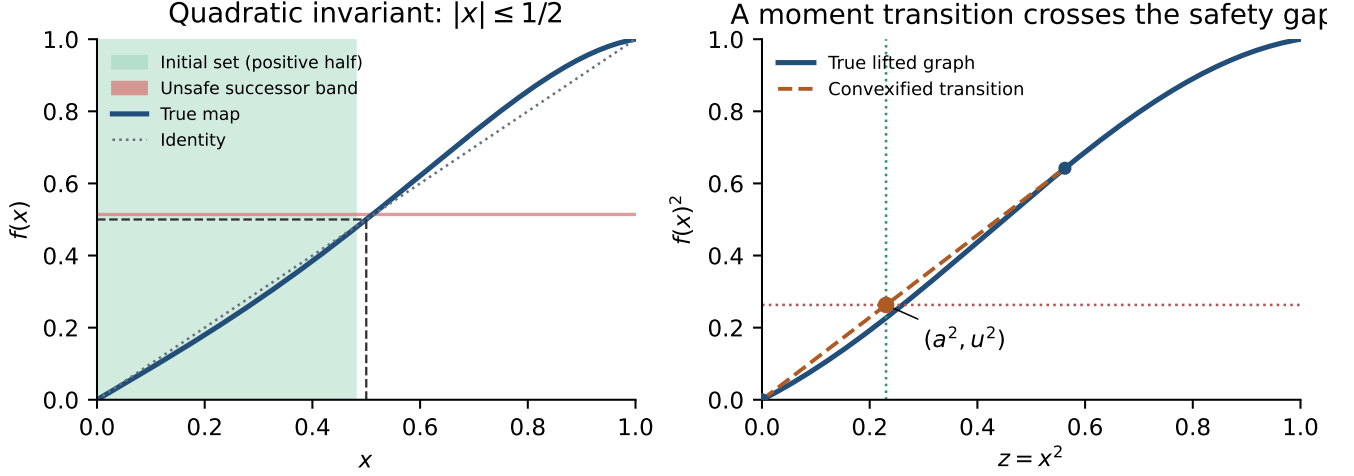


Figure. Positive half of the symmetric example. In the second-moment plane the chord from $(0,0)$ to $(t^2, f(t)^2)$ contains (a^2, u^2) . It connects realizable initial moments to realizable unsafe moments, although the true system keeps $[-1/2, 1/2]$ invariant.

Verified computation. `src/exact_verify.py` uses only Python’s standard-library rational arithmetic. It verifies support and probability constraints, every moment through degree three, the invariant-domain derivative identity, all margins, and the exact factorization and Bernstein coefficients in (3). Five evidence tests include rejection of a corrupted successor witness and rejection of the same atoms as a degree-four witness. A separate finite-support LP discovery routine recovers a degree-three witness, reconstructs its weights rationally, and repeats exact replay. Its degree-four grid is infeasible; that solver status is *not* the proof of minimum degree. The analytical obstruction plus the explicit quartic certificate establish the degree claim. Plotting is illustrative only. No SOS synthesis run is reported.

Closest work	Established contribution	Distinction / overlap to assess
Submitted VBC–IBC paper [1]	Affine convexified transition obstruction; path/scaled duality; rotation separation.	Moment lift extends the obstruction to prescribed degree; exact degree-two versus degree-four example.
Horvath–Song–Terlaky [2]	Theorem 3.1 gives global nonnegative comparisons for affine polyhedral inequalities when successor constraints are concave.	Theorem 1 uses convexification in certificate-value space for arbitrary continuous B . Its duality mechanism is classical and should not alone be claimed as the principal novelty.
Bogomolov et al. [3]	Convex interpolation discovers template directions that eliminate spurious hybrid counterexamples.	A moment witness identifies spurious transitions that <i>no added degree-d inequality</i> can eliminate under global comparison. Models and propagation rules differ.
Anand et al. [4]	Path-completeness for soundness and graph ordering via simulation.	Ordering graph structures is distinct from changing global comparisons to sign implications. No switched completeness theorem is claimed here.
Korda–Henrion–Jones [5]	Occupation-measure LPs and moment/SOS hierarchies for invariant sets.	The tools are established; the proposed target is nonexistence for an entire finite-degree VBC class with unrestricted finite vector size.

Domain caveat in the invariance reference. Horvath et al.’s Theorem 5.1 imposes its comparison only *on the invariant polyhedron*; choosing the zero matrix then proves necessity. It does not imply a comparison on the larger verification domain X . This difference is central here. Their Theorem 3.1 does require a global inequality and additional concavity assumptions, consistent with the Jensen regime on page 2.

6 Novelty assessment and the next TAC-level question

What is established in this draft. Three explicit statements have proofs: a fixed-vector global-comparison characterization, a component-count-independent sufficient degree obstruction, and a parameterized exact 2–4 minimum-degree gap. The first target in Majid’s email is met more strongly than requested *within the fixed-anchor class*. These are proposed results with checked finite evidence; they still require independent mathematical and bibliographic review. The present search is targeted, not exhaustive, and does not certify priority or TAC suitability.

What the attached publication contributes. The path-complete closure-certificate paper [6] proves a graph soundness characterization for persistence using ordered node-pair relations, with a separate reachability–rank construction. It motivates treating propagation structure as a design choice. It does not supply a fixed-polynomial-degree completeness result, and its relational certificates and persistence specification should not be silently identified with the autonomous safety barriers studied here.

Recommended core for a TAC paper. Develop a *degree-preserving formulation diagnosis*: determine when refinement by more polynomial inequalities can succeed in the convexified feature-transition relation, and when changing the propagation rule is required. The compelling contribution would combine an existence/characterization theorem, exact counterexamples, and a reproducible diagnosis algorithm. The convex-duality theorem and one engineered example alone may be insufficient for TAC.

1. **Extend the obstruction through multiple lifted steps.** If $(z_{j-1}, z_j) \in F_d$ for $j = 1, \dots, \ell$, with $z_0 \in K_{0,d}$ and $z_\ell \in K_{u,d}$, then every forward degree- d VBC obeys $Cz_\ell \leq A^\ell Cz_0 \leq 0$, a contradiction. The backward proof reverses the inequalities. This is a proved sufficient extension, not yet a complete decision method.
2. **Seek a converse under robust assumptions.** Establish conditions under which safety of the lifted convex relation yields a finitely faceted invariant feature polyhedron with strict initial inclusion and a fixed unsafe anchor. The existence interpretation on page 2 would then convert it into a VBC. Compactness alone is not asserted to ensure such a finite polyhedron or a terminating algorithm.
3. **Use counterexamples to select formulations.** First search for exact atomic obstructions at the requested degree. A verified obstruction justifies switching to implication conditions or increasing degree. Otherwise continue synthesis or refine supports; absence of a discovered witness is inconclusive. For fixed B , search for a violating probability mixture using Theorem 1, then refine B or change propagation conditions.
4. **Broaden examples with controlled scope.** The scalar family embeds into $X = [-1, 1] \times [-1, 1]^q$ with transverse contraction and initial/unsafe product sets. Restricting any degree-three multivariate VBC to the invariant axis $y = 0$ inherits the obstruction. Coupling can be introduced via $c(y)$ in the permitted parameter range. These are structured embeddings, not evidence of application-scale performance.

Computational safeguards. A feasible moment-SOS relaxation can contain nonrepresentable pseudomoments; it cannot establish VBC nonexistence without extracted, support-verified measures or another valid analytical argument. Finite-grid LP infeasibility does not exclude witnesses on other supports. Unknown A and unknown B produce bilinear synthesis constraints, so a fixed- A SDP is not a complete search. A positive solver result requires exact replay; failures remain failures of that search. The open parameter result establishes a family, not robustness to arbitrary perturbations of dynamics.

Proposed coauthor decision. Check the proof and scope of Theorems 1–3 and assess the exact degree gap against existing polynomial-template lower bounds. If these survive review, prioritize the robust converse and the multistep diagnosis algorithm before adding switching, control, data-driven guarantees, or persistence specifications.

References

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